🧾 Intermediate Report — Urdu Toxic Comment Detection (NLP Project)

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1. Project Overview

The project aims to develop an **automated toxicity detection system** for **Urdu and Roman-Urdu social-media comments**. The goal is to classify comments as **toxic (1)** or **non-toxic (0)**, capturing explicit abuse, hate speech, and implicit toxicity.

The system integrates three publicly available datasets: *RUT*, *RUHSOLD*, and *Urdu YouTube*, to build a combined multilingual corpus, followed by baseline modeling and lexicon sensitivity analysis.

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2. Data Preprocessing

- **Datasets used:**
    - **RUT Dataset (72 k)** – Roman Urdu tweets/comments with toxicity labels
    - **RUHSOLD Dataset (10 k)** – Hate-speech corpus with explicit slurs and `hate_words_roman_urdu.txt` lexicon
    - **Urdu YouTube Dataset (2.1 k)** – Manually extracted comments in Urdu script
- **Final merged corpus: 84,954 rows**, 4 columns (`text`, `label`, `source`, `script`)
- **Preprocessing steps:**
    - Unicode normalization, HTML tag & emoji removal
    - Lowercasing + punctuation cleaning
    - Script tagging: *roman* vs *urdu*
    - Label harmonization (`TRUE/FALSE` → 1/0)

- Tokenization and vectorization using **TF-IDF (1-2 grams, max_features = 25 k)**

✅ *Balanced sampling and consistent cleaning were applied across datasets to ensure comparable feature representation.*

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3. Exploratory Data Analysis (EDA)

Analysis	Key Observation
Class Distribution	77 % non-toxic vs 23 % toxic → moderate imbalance
Source Distribution	RUT dominates (72 k); RUHSOLD (10 k); YouTube (2 k)
Toxic vs Non-Toxic by Source	RUT mainly clean; RUHSOLD balanced; YouTube slightly toxic
Script Distribution	82 k Roman Urdu vs 2 k Urdu script
Comment Length Histogram	Median ≈ 100 chars; long-tail ≤ 400 chars
Top Words	Common tokens: *hai*, *ki*, *ka*, *to* → Stop-word dominance → lexical toxicity ≠ semantic toxicity

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4. Baseline Model Results

Model: Logistic Regression (TF-IDF, class_weight = 'balanced')
Train/Test Split: 80 / 20
Performance (overall):

Metric	Non-Toxic (0)	Toxic (1)
Precision	0.94	0.62
Recall	0.85	0.82
F1-score	0.89	0.71

Accuracy: 0.845 **Macro F1:** 0.80

By Script: Roman → F1 ≈ 0.70 | Urdu → F1 ≈ 0.91
By Source: RUT (0.74) | RUHSOLD (0.59) | YouTube (0.92)

📘 *Interpretation:* The baseline model captures explicit abuse well but struggles with context-dependent or subtle toxicity in Roman-Urdu text.

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5. Lexicon Sensitivity Analysis

Objective: Quantify reliance on explicit hate-word lexicon.
Approach: Mask known slurs with `[insult]`, re-evaluate model probabilities.

Metric	Result
Sample size (lexicon-containing rows)	8 000
Toxic → Non-Toxic flips	375 (4.7 %)
Non-Toxic → Toxic flips	172 (2.1 %)
Mean ΔP(toxic)	− 0.002
Highest flip source	RUHSOLD (11.3 %) > RUT (3.6 %)

Findings:

- Around 5 % of predictions flipped, proving the model relies moderately on explicit lexical cues.
- RUHSOLD samples were most affected → hand-crafted slurs dominate toxicity cues.
- Masking reduced P(toxic) slightly (mean Δ = −0.002) but did not destabilize predictions.

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6. Key Insights & Limitations

- **Lexical bias:** Model performance depends on surface-level slurs; fails for implicit hate or sarcasm.
- **Domain bias:** RUT domination reduces generalization to formal or Urdu-script data.
- **Class imbalance:** Toxic minority class still under-represented despite weighting.

- **Feature limitation:** Bag-of-Words features ignore context, sentiment, and code-switching.

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7. Next Steps

Stage	Plan	Expected Outcome
Model Upgrade	Fine-tune **Multilingual BERT (mBERT / XLM-R)** on the combined corpus (Roman + Urdu).	Context-aware embeddings capturing implicit toxicity
Augmentation	Use synonym replacement + back-translation to enlarge toxic minority samples.	Reduced class imbalance
Lexicon Integration	Incorporate lexicon features as auxiliary input to transformer model.	Hybrid lexical-semantic detection
Evaluation	Stratified testing by script & source; macro-F1 tracking.	Fairer cross-domain comparison

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